Nigus Teklehaymanot

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Profile

Ph.D. Candidate in Mathematics at the University of California, Irvine, with fourteen years of college mathematics teaching experience in California community colleges and at universities in Ethiopia, and as a teaching assistant at UC Irvine since Fall 2023. I have taught developmental through advanced undergraduate courses, including calculus, linear algebra, differential equations, probability and statistics, number theory, and optimization, and I have advised undergraduate senior seminar projects.

Education

University of California, Irvine — Irvine, CA
Ph.D. Candidate, Mathematics

Norwegian University of Science and Technology (NTNU) — Trondheim, Norway
M.S. in Mathematical Sciences — May 2018

Mekelle University — Mekelle, Ethiopia
B.S. in Mathematics and Physics — July 2008

Samara University — Samara, Ethiopia
Licensed Professional Teacher Educator — May 2010

Teaching Experience

Teaching Assistant — Fall 2023–Present

University of California, Irvine

- Teaching assistant for Calculus I, Calculus II, Multivariable Calculus I, Multivariable Calculus II, Introduction to Linear Algebra, Elementary Differential Equations, Numerical Analysis I, Numerical Analysis II, Introduction to Partial Differential Equations, Linear Algebra I, and Linear Algebra II.

Adjunct Instructor of Mathematics — July 2019–Present

Laney College, Oakland, CA

- Teach Pre-Calculus, Calculus I, Introduction to Statistics, Intermediate Algebra, and Math for Lab Science.

Adjunct Instructor of Mathematics — July 2021–Present

Contra Costa College, San Pablo, CA

- Teach Introduction to Probability and Statistics and Intermediate Algebra.

Adjunct Instructor of Mathematics — September 2019–Summer 2022

Diablo Valley College, Pleasant Hill, CA

- Taught College Algebra, Statway I, Calculus I, and Calculus II.

Lecturer of Mathematics — September 2013–August 2015

Mekelle University, Mekelle, Ethiopia

- Taught Number Theory, Linear Algebra, Calculus III, and Optimization Theory.
- Served as a formal adviser to undergraduate students on senior seminar projects.

Assistant Lecturer of Mathematics — September 2008–August 2013

Samara University, Samara, Ethiopia

- Taught Calculus I, Calculus II, Solid Geometry, and Differential Equations.

Coordinator of Math Club and Tutoring Service — February 2009–August 2010

Samara University, Samara, Ethiopia

- Organized a mathematics competition for undergraduates and arranged tutoring services.

Courses Taught

Lower-division and supporting courses: Intermediate Algebra; College Algebra; Pre-Calculus; Calculus I; Calculus II; Multivariable Calculus; Introduction to Probability and Statistics; Introduction to Statistics; Statway I; Math for Lab Science.
Additional undergraduate courses: Calculus III; Linear Algebra; Differential Equations; Partial Differential Equations; Numerical Analysis; Number Theory; Optimization Theory; Solid Geometry.

Instructional Leadership and Service

- Curriculum development committee member, Mekelle University, 2014.
- Research and module publication committee member, Samara University, 2010.
- Curriculum development committee member, Samara University, 2009.
- Volunteer mathematics tutor for undergraduate summer students, Mekelle University, 2008.

Honors

Merit-based full scholarship, Norwegian Quota Scholarship Scheme, Norway, 2015.

Research and Scholarly Work

Doctoral research in tensor methods and their applications, including the t-product, t-SVD, and tensor robust PCA. Advancement paper: *Computing the t-SVD of a General Third-Order Tensor Product and Quotient.*

Manuscripts in progress

- *Multi-Look Imaging of Moving Targets in Clutter with a Stationary MIMO Radar via t-SVD.*
- *Line-of-Sight Diversity for Moving-Target Separation in Wide-Angle and Circular SAR via Robust PCA.*
- *Ego-Motion-Aware Tensor Robust PCA for Static Clutter Suppression and Moving Target Detection in MIMO Radar.*

Collaborative papers with faculty advisor

- *Solving Multilinear Systems via Tensor Inversion under the t-Product Framework: Theory, Algorithms, and Applications.*
- *Backward Error, Perturbation Bounds, and Efficient Solvers for the t-Sylvester Tensor Equation.*

Research Assistant September 2016–December 2017

NTNU, Norway

- Assisted Prof. Kristian Seip on "Dirichlet Series and Analysis on Polydiscs," contributing findings related to final thesis work.

Journal Publications

- *Balance an Unbalanced Transportation Problem by a Heuristic Approach.* Ethiopia, 2013. Co-authored and proofread with Dr. Tripti Sharma.
- *Minimizing the Range of Lateness for n-Jobs, 3-Machines Sequencing Problem with Due Date, Transportation Times and Equivalent Job for Block of Jobs – A Heuristic Approach.* Ethiopia, 2015. Co-authored and proofread with Dr. Tripti Sharma.

Skills and Languages

Teaching: Course instruction from developmental mathematics through advanced undergraduate topics; senior-seminar advising; tutor supervision; curriculum committee service; research and report writing.
Computational tools: MATLAB, Python, R, Tableau, SQL, SAS, STATA; Office Suite.
Languages: English (Proficient); Tigrigna (Native); Amharic (Fluent); Norwegian (Basic).